Megan R. Lancaster

Senior Learning Experience Designer, STEM Educator, and Project Manager

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Summary

Science educator, learning experience designer, and PMP®-certified project manager with 13 years building rigorous, technology-enabled learning across higher education, K-12 STEM, and professional education. Background spans physics, engineering physics, science education, and science education research. At Duke, leads end-to-end design and delivery for 50+ online courses and Specializations across engineering, data science, healthcare, and business; builds AI-assisted workflows and interactive learning tools; and uses learner feedback and analytics to iterate. Brings academic rigor, product-minded design, and a practical approach to building, testing, and shipping accessible learning experiences.

Professional Experience

Senior Learning Experience Designer and Project Manager

Duke University | Durham, NC (Remote) | April 2021 - Present

- Lead learning design and project delivery for 50+ online courses and Specializations across engineering, data science, healthcare, and business, translating complex expert content into coherent course structures, applied practice, and assessment.
- Guide multidisciplinary teams from early discovery through launch, defining scope, coordinating resources and reviews, adapting plans as requirements change, and helping stakeholders make decisions about quality, timing, and capacity.
- Design hands-on coding practice, branching scenarios, interactive media, and case-based learning; use learner feedback, expert consultation, prototypes, and analytics to refine experiences around what learners need to understand and do.
- Designed and built a Claude API-powered Interactivity Generator and Coursera plugin that combines activity selection, working previews, content editing, expert review, and export in one workflow, with static-text alternatives for learners.
- Co-designed Coursera's integrated labs interface, now deployed platform-wide, applying Universal Design for Learning to make technical learning environments easier to navigate.
- Created reusable course design resources used in 25 published courses and trained faculty, Duke Corporate Education partners, and fellow designers to use them; strengthened delivery through scoping tools, resource plans, quality-review frameworks, and AI-assisted workflows.

Learning Experience Designer

Duke University | Durham, NC (Remote) | January 2020 - April 2021

- Led design sprints with faculty across Duke University and Duke Health, translating complex subject matter into online and hybrid learning experiences.
- Developed interactive prototypes and assessments; used learner analytics and faculty feedback to refine course designs.

Educational Technology Specialist

High Point University | High Point, NC | July 2018 - December 2019

- Led campus-wide learning technology initiatives supporting 100+ faculty; created and scaled Blackboard Bootcamp, improving faculty confidence by 32% and reducing support requests by an average of 18%.

- Designed the university's first cross-disciplinary Pathway system using e-portfolios and achievement badges, giving leadership greater visibility into student progression; analyzed adoption and engagement data to guide improvement.

Program Manager, K-5 Engineering Education

UNC Greensboro | Greensboro, NC | August 2015 - December 2017

- Coordinated Engineering is Elementary implementation across schools, coached teachers, adapted programming for different learning contexts, and developed mixed-methods evaluation instruments to inform program and district planning.

Instructional Designer and Project Manager

UNC Greensboro | Greensboro, NC | September 2013 - July 2015

- Managed online course development teams of faculty, multimedia developers, and designers; designed and launched UNCG's first MOOC and supported undergraduate and graduate course development across disciplines.
- Coordinated milestones, accessibility requirements, and stakeholder needs for courses built in Canvas and WordPress.

Selected STEM Curriculum, Teaching & Research

- **Virtual Mission Flight Director, Challenger Learning Center (2024-2025)**: led virtual Moon and Mars missions through a NASA educator program, prepared supporting curriculum with K-12 teachers, and provided live technical support to participating teachers and students.
- **Science Teacher, New Garden Friends School (2014-2015)**: designed and taught middle school science and high school environmental science using hands-on, guided inquiry and Next Generation Science Standards (NGSS).
- **STEM Instructor and eStudies Academic Consultant, Duke TIP (2011-2015)**: designed and taught courses including Roller Coaster Physics, Engineering of Space Exploration, Digital Circuit Design and Analysis, and Engineering Technology for the Digital World; reviewed online STEM curriculum and trained instructional staff.
- **Graduate Teaching Assistant, NC State and UNC Chapel Hill (2011-2013)**: supported Methods of Teaching Science, Teaching and Learning Physics, General Physics II recitation, and Electronics I Lab.
- **Science and engineering education research** included student and teacher interviews, classroom observation, research-protocol design, and mixed-methods analysis; earlier research included stellar evolution / absorption-spectra analysis and mathematical biology modeling.

Education & Certification

Project Management Professional (PMP), Project Management Institute
M.Ed., Science Education, NC State University, STEM Education Research and Curriculum Development
M.S., Engineering Physics, Appalachian State University, Instrumentation and Automation
B.S., Physics, University of North Carolina at Greensboro

Tools & Recognition

AI and development: Claude, Claude API, Claude Code, Claude Design, ChatGPT; AI-assisted HTML/JavaScript development
Platforms and authoring: Coursera, Canvas, Blackboard, WordPress, PlayPosit, Monday.com; Articulate Storyline, Vyond, Camtasia, Adobe Captivate, Illustrator, Character Animator
Team Recognition: Coursera Vision & Values Award (2025); Duke Presidential Award (2021)
Scholarship: Peer-reviewed publications in the Journal of Research in Science Teaching and Science Scope; presentations at NARST, AERA, UPCEA DT&L, and Coursera Conference